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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Harshal Mohadikar
(11/08/2026)
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2026-27 (ODD)	Course Name	Lab : Generative AI
Semester	7	Course Code	23ADS1706
Roll No	41	Name of Student	Krishna Mishra

Practical Number	3
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO 2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5 to solve real-world AI and NLP problems.. CO 3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains.. CO 4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications..
Aim	Next Word Prediction using LSTM
Problem Definition	Develop a predictive typing system for a mobile keyboard that suggests the next word while typing.
Theory (100 words)	Long Short-Term Memory (LSTM) networks are a specialized type of Recurrent Neural Network (RNN) designed to capture long-term dependencies in sequential data. Unlike standard RNNs, LSTMs use memory cells and gating mechanisms (input, forget, and output gates) to selectively retain or discard information over time, effectively mitigating the vanishing gradient problem. For next word prediction, the model takes a sequence of previous words as input and outputs a probability distribution over the vocabulary for the next word. The Embedding layer converts words to dense vectors, LSTM layers capture sequential context, and a Dense layer with Softmax activation produces the final prediction probabilities.

Procedure and Execution (100 Words)	Algorithm: 1. Imported required TensorFlow/Keras layers (Embedding, LSTM, Dense) and utilities (Tokenizer, pad_sequences) 2. Defined sample text data containing four sentences about next word prediction and LSTMs. 3. Created a Tokenizer to fit on the text, building a word-to-index mapping with a total vocabulary size of 39 words. 4. Generated input sequences by creating n-grams from each sentence, where each sequence consisted of all previous words as predictors and the next word as the label. 5. Padded all sequences to ensure uniform length and converted labels to one-hot encoded vectors. 6. Built a Sequential model with Embedding (100 dimensions), two LSTM layers (150 and 100 units), and a Dense output layer with Softmax activation. 7. Compiled the model with categorical cross-entropy loss and Adam optimizer, then trained for 100 epochs.

Code:

```
import numpy as np
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense

# Sample data for the predictive typing system
data = """Next word prediction is a critical task in natural language processing.
LSTMs are great for sequence prediction tasks like this.
Developing a mobile keyboard requires efficient models.
Deep learning helps in suggesting the most probable next word."""

# Tokenization
tokenizer = Tokenizer()
tokenizer.fit_on_texts([data])
total_words = len(tokenizer.word_index) + 1

# Create input sequences
input_sequences = []
for line in data.split('\n'):
    token_list = tokenizer.texts_to_sequences([line])[0]
    for i in range(1, len(token_list)):
        n_gram_sequence = token_list[:i+1]
        input_sequences.append(n_gram_sequence)

# Pad sequences
max_sequence_len = max([len(x) for x in input_sequences])
input_sequences = np.array(pad_sequences(input_sequences, maxlen=max_sequence_len, padding='pre'))

# Create predictors and label
X, y = input_sequences[:,:-1], input_sequences[:,-1]
y = tf.keras.utils.to_categorical(y, num_classes=total_words)

print(f'Total words: {total_words}')
print(f'Max sequence length: {max_sequence_len}')
```

	`` # Define the LSTM model model = Sequential([Embedding(total_words, 100, input_length=max_sequence_len-1), LSTM(150, return_sequences=True), LSTM(100), Dense(total_words, activation='softmax')]) model.compile(loss='categorical_crossentropy', optimizer='adam', metrics=['accuracy']) model.summary() # Train the model history = model.fit(X, y, epochs=100, verbose=1) def predict_next_word(seed_text): token_list = tokenizer.texts_to_sequences([seed_text])[0] token_list = pad_sequences([token_list], maxlen=max_sequence_len-1, padding='pre') predicted = np.argmax(model.predict(token_list, verbose=0), axis=-1) output_word = "" for word, index in tokenizer.word_index.items(): if index == predicted: output_word = word break return output_word # Test the prediction seed = "next word" prediction = predict_next_word(seed) print(f"Input: '{seed}'") print(f"Suggested next word: '{prediction}'") ```
Output Analysis	The model was successfully trained on the sample text, achieving increasing accuracy over the 100 training epochs as the LSTM learned sequential patterns. The total vocabulary size was 39 words, with a maximum sequence length of 12 words. When tested with the seed text "next word," the model suggested the word "prediction" as the next word, which is contextually appropriate given that the training data frequently contained the phrase "next word prediction." This demonstrates the LSTM's ability to capture sequential relationships and generate meaningful predictions based on learned patterns.
Link of student Github profile where lab assignment has been uploaded.	krish-na45/GenAI-Practicals

Plag Report (Similarity index < 12%)	
Conclusion	The LSTM-based model effectively demonstrates next word prediction capabilities, successfully suggesting contextually relevant words based on input sequences. This approach forms the foundation of modern predictive typing systems, showing that even with limited training data, LSTMs can capture meaningful sequential patterns and provide useful word suggestions for mobile keyboards.
Date	11/08/2026